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A new perspective on gold as a risk hedge: Long-term impacts of bilateral political tensions between the U.S. and China

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ABSTRACT

Based on the GARCH-MIDAS model, this study systematically examines the impact of U.S.-China geopolitical tensions on the volatility of the spot gold market. By introducing a monthly U.S.-China tension index (UCT), we quantify the explanatory power of bilateral geopolitical friction on long-term volatility in the gold market. Empirical results indicate that the gold market exhibits significant volatility clustering, and the UCT significantly amplifies long-term volatility. This reflects the fact that rising geopolitical tensions trigger risk-averse behavior in the market, driving capital reallocation toward safe-haven assets, such as gold. After incorporating monthly realized volatility (RV) as a control variable, the UCT remains statistically significant, further validating its independent information-additive value. Out-of-sample forecasting results indicate that the two-factor model, including UCT and RV, outperforms the single-factor model in volatility forecasting, with stronger predictive capability and greater stability. This study validates the need to incorporate the gold market into a systemic risk monitoring and early warning system centered on geopolitical risk. It proposes it as an important reference indicator for macroprudential regulation.

1. Introduction

Against the backdrop of turmoil in the global governance system and the continued rise in market risk aversion, the importance of gold as a traditional safe-haven asset has become increasingly evident, as its price has risen steadily in recent years. Extensive research has shown that during periods of heightened economic and policy uncertainty, gold not only fulfills its traditional safe-haven function but also serves multiple roles, including wealth preservation and hedging against systemic risk (Baur & McDermott, 2010). First, from the perspective of the store-of-value theory, gold is widely regarded as a crucial tool for mitigating inflation and credit risks. Existing research reveals that gold prices exhibit long-term cointegration with the money supply, inflation expectations, and the real exchange rate of the US dollar, indicating that its "monetary anchor" property remains highly effective in contemporary times (Ghosh et al., 2004). Second, grounded in uncertainty aversion and safe-haven theories, gold serves as a natural hedge against systemic financial risks, substantially enhancing portfolio stability during severe market downturns or extreme volatility (Baur & Lucey, 2010). Overall, when financial markets experience extreme turbulence or macroeconomic uncertainty surges, investors tend to increase gold holdings to mitigate potential losses, thereby driving up market demand and prices.

As geopolitical conflicts become increasingly frequent, the interplay between gold and political risks has drawn widespread

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attention. Geopolitical risks not only erode confidence in fiat currencies but also amplify volatility in the gold market by disrupting global supply chains and international capital flows. During peacetime, gold exhibits low correlation with equity markets (such as the S&P 500 Index). However, amid major political events, their correlation increases significantly, highlighting gold's diversification and hedging value in high-risk environments. It particularly demonstrates robust safe-haven asset attributes during wartime scenarios (Baur & Smales, 2020; Pastén-Henríquez et al., 2025; Triki & Ben Maatoug, 2021).

The impact of geopolitical risks on gold extends beyond yield levels, manifesting more prominently as heightened volatility. An integrated analysis of geopolitical risk indicators across multiple countries reveals a significant positive correlation between political risk levels and gold volatility, with predictive power further enhanced through dimensionality reduction (Li et al., 2023). During periods of extreme political events, precious metals such as gold typically exhibit consistent safe-haven characteristics (Baur & Smales, 2020).

The intensity and persistence of geopolitical events' impact on the gold market vary significantly across different event types. Actual wars or terrorist attacks are more likely to trigger gold price bubbles and extreme tail risks than events remaining at the threat stage (Gupta et al., 2017; Zhou & Liang, 2025). Furthermore, during heightened geopolitical instability, central banks tend to substantially increase gold holdings to hedge against potential financial sanctions and monetary system risks. This official reserve behavior provides additional medium-to long-term support for gold prices (Ngo et al., 2024).

Furthermore, geopolitical risks do not isolate the gold market; they often influence macroeconomic variables (such as inflation expectations, interest rates, and exchange rates) and financial market structures, forming multi-channel, cross-temporal risk transmission mechanisms. Existing research indicates that major geopolitical events can transmit emerging market risks to the gold market through "interconnected spillover effects", significantly amplifying volatility during the initial stages of such events (Li et al., 2021). Consequently, gold price fluctuations are influenced not only by direct shocks but also by indirect factors such as cross-asset flows, cross-border capital migration, and shifts in monetary policy expectations. Against the backdrop of persistently escalating global political and economic uncertainties, gold has evolved from a traditional safe-haven asset into a comprehensive financial instrument that combines hedging, asset allocation, and strategic reserve functions.

This study systematically reviews the literature on the impact of geopolitical risks on gold market volatility and finds that most scholars focus on the Global Geopolitical Risk Index (GPR) proposed by Caldara and Iacoviello (2022). While such indicators can capture the overall trend of global political turbulence, they struggle to identify structural shocks triggered by specific bilateral strategic relationships, particularly the strategic friction between the United States and China. As the world's two largest economies, the evolving relationship between the U.S. and China has significant spillover effects on global financial markets. Rogers et al. (2024) constructed the "U.S.–China Tension Index (UCT)" based on news corpora. The UCT indicator directly measures the level of strategic tension between the United States and China, capturing granular policy risks, including trade conflicts, technology controls, military frictions, and diplomatic disputes. Such "structured, directional bilateral political conflicts" often correspond to clear shifts in global capital flows and heightened risk-aversion demands, exhibiting an information structure markedly different from traditional GPR.

Based on this, this study introduces the monthly UCT index as the core explanatory variable. It employs the GARCH-MIDAS model to assess its impact on the volatility of daily log returns in the gold spot market. The study found that the θ coefficient for the UCT is statistically significant and positive, indicating that rising U.S.-China geopolitical tensions will significantly increase the long-term volatility of gold. More importantly, even after controlling for monthly realized volatility (RV), UCT still exhibits significant incremental explanatory power, suggesting that the information it carries goes beyond traditional historical volatility indicators.

Compared with existing research, this study advances the literature through model design, variable construction, and empirical evaluation. First, drawing on the GARCH-MIDAS framework of Engle et al. (2013), it incorporates the U.S.–China Tension Index (UCT) to identify the long-term impact of bilateral geopolitical frictions on gold market volatility. This approach fills an important gap in the literature, which has largely overlooked how specific and strategically significant geopolitical dyads shape volatility dynamics in safe-haven assets. Second, the empirical results show that UCT exerts a statistically significant and robust effect on long-term volatility. Even after controlling for monthly realized volatility, UCT retains strong explanatory power, demonstrating that it provides independent and incremental geopolitical information beyond high-frequency market fluctuations. This finding not only validates UCT as an effective geopolitical risk measure for the precious-metals market but also deepens understanding of the transmission mechanism linking geopolitical tensions and safe-haven demand. Finally, through out-of-sample forecasting and a series of robustness tests, this study confirms that the $GARCH-MIDAS_{RV+UCT}$ model delivers superior predictive performance relative to benchmark models across multiple loss functions. These results highlight the model's practical value and underscore the importance of incorporating geopolitical tensions into long-horizon volatility forecasting.

In light of these results, this study recommends that regulatory authorities should enhance their strategic understanding of the gold market, regard it as an important indicator of systemic risk, and establish a dynamic financial early warning system centered on geopolitical risk in policy practice. They should also optimize the trading mechanisms and information disclosure mechanisms of the gold market. Additionally, price fluctuations in commodities, including gold, should be incorporated into the macroprudential regulatory framework to identify better and manage potential systemic financial risks.

2. Methodology

2.1. GARCH-MIDAS model

This study uses the GARCH-MIDAS model proposed by Engle et al. (2013) to address the time-frequency mismatch between high-frequency daily gold volatility and low-frequency monthly UCT. This model provides a solid methodological foundation for the

empirical analysis in this study.

$$r_{i,t} = \mu + \sqrt{\tau_t * g_{i,t}}\varepsilon_{i,t} \ \forall \mathrm{i} = 1, \ldots, \mathrm{N_t} \tag{1}$$

In Formula (1), μ denotes the unconditional mean, the subscript t represents the low-frequency time dimension in months, and $r_{i,t}$ denotes the daily log return of gold spot prices. This model decomposes the conditional variance into two components: the short-term volatility term $g_{i,t}$ and the long-term volatility term τ_t, which respectively reflect the short-term dynamic characteristics of high-frequency data and the long-term impact of low-frequency macroeconomic factors on volatility.

$$g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{\left(r_{i-1,t} - \mu\right)^2}{\tau_t} + \beta g_{i-1,t} \tag{2}$$

In formula (2), the short-term volatility term $g_{i,t}$ is modeled using a GARCH (1,1) model to characterize the high-frequency volatility characteristics of daily gold returns. To ensure the validity and stability of the model, the parameters must satisfy the following conditions: $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta < 1$, thereby ensuring the positivity of the conditional variance and cointegration stability.

$$ln\ \tau_t = m + \theta_X \sum_{k=1}^{K} \varphi_k(w_1, w_2) X_{t-k} \tag{3}$$

In equation (3), the long-term volatility term, cap X, is composed of the realized volatility and the rate of change of the UCT; its logarithm is taken to ensure that the long-term volatility term, τ_t is always positive. Based on the GARCH-MIDAS model framework, this study extends equation (3) to construct a two-factor GARCH-MIDAS model that simultaneously incorporates two low-frequency variables. The core objective of this setting is to investigate whether the UCT has independent and significant predictive power for explaining the long-term volatility of the gold market, after controlling for market status information represented by realized volatility.

As Chauvet et al. (2015) pointed out, realized volatility typically reflects the market's high sensitivity and immediate response to current macroeconomic conditions. Therefore, if a variable significantly improves volatility predictability after controlling for realized volatility, it indicates that it contains incremental information beyond current market conditions, demonstrating strong forward-looking and systemic risk disclosure capabilities. The two-factor model was designed precisely to determine whether UCT possesses such independent informational value.

$$ln\ \tau_t = m + \theta_{RV} \sum_{k=1}^{K} \varphi_k(w_1, w_2) RV_{t-k} + \theta_{UCT} \sum_{k=1}^{K} \varphi_k(w_1, w_2) UCT_{t-k} \tag{4}$$

In Equation (4), K represents the maximum lag order of long-term fluctuations. According to Engle et al. (2013), when the model involves variables with mixed high-frequency (daily) and low-frequency (monthly) frequencies, setting K to 36 achieves a better balance between model complexity and forecasting accuracy. Therefore, this study also adopts $K = 36$ in the empirical analysis to ensure the comparability and robustness of the model's forecasting performance.

Additionally, realized volatility is calculated as $RV_{t-k} = \sum_{k=1}^{N_t} r_{t-k}^2$, which is the sum of the squares of the daily log returns of gold spot prices, yielding the realized volatility for the corresponding month.

In Equations (3) and (4), this study primarily focuses on changes in the long-term coefficient theta to assess the explanatory power of different low-frequency variables for the long-term volatility of the gold market. The weighting function $\varphi_k(w_1, w_2)$ is a Beta distribution function, whose parameters w_1 and w_2 control the weight distribution of each lagged variable, used to characterize the dynamic influence path of long-term information on volatility.

$$\varphi_k(w_1, w_2) = \frac{\left(\frac{k}{K}\right)^{w_1 - 1}\left(1 - \frac{k}{K}\right)^{w_2 - 1}}{\sum_{j=1}^{K}\left(\frac{j}{K}\right)^{w_1 - 1}\left(1 - \frac{j}{K}\right)^{w_2 - 1}} \tag{5}$$

The Beta weighting function in Formula (5) characterizes the lagged impact of low-frequency variables on high-frequency volatility and their diminishing effect. To maintain the model's rationality, this study also sets $w_1 = 1$ to ensure that the impact of exogenous variables on long-term volatility exhibits a monotonic decreasing trend.

The specific form is shown in Formula (6):

$$\varphi_k(w_2) = \frac{\left(1 - \frac{k}{K}\right)^{w_2 - 1}}{\sum_{j=1}^{K}\left(1 - \frac{j}{K}\right)^{w_2 - 1}} \tag{6}$$

Therefore, equations (1)–(3) and (6) together constitute the complete form of the GARCH-MIDAS single-factor model; while equations (1), (2), (4) and (6) constitute the extended setting of the GARCH-MIDAS dual-factor model. The former is used to analyze the impact of a single low-frequency variable on long-term volatility, while the latter further tests the marginal explanatory power of the UCT, controlling for realized volatility.

2.2. Model evaluation methods

To maximize the reliability of model evaluation results, this study employs three widely used methods for evaluating out-of-sample prediction performance. First, the Model Confidence Set (MCS) method, proposed by Hansen et al. (2011), is used to systematically screen out-of-sample prediction performance and identify prediction schemes that are statistically superior to other models. Second, the out-of-sample R^2 metric proposed by Rapach et al. (2010) is used to assess whether the predictive capability of the constructed model in the out-of-sample data exceeds that of the benchmark model. Finally, the DM test proposed by Diebold and Mariano (2002) is employed to assess the significance of the difference between the predictive and benchmark models in terms of reducing prediction error.

2.2.1. Model Confidence Set (MCS) test

This study adopts the MCS testing method proposed by Hansen et al. (2011). One of the major advantages of this method is that it does not require the prior specification of a specific benchmark model. Instead, based on the prediction loss function, it gradually excludes models that perform significantly worse, at a specified confidence level (p = 0.10), ultimately retaining a confidence set of models with relatively superior predictive performance.

Given that Diebold and Lopez (1996) noted that it is challenging to define the "optimal" loss function for assessing the accuracy of volatility prediction, this study selects three commonly used loss function metrics from the literature to comprehensively measure the prediction error of the models. These include the mean squared error (MSE), mean absolute error (MAE), and log-log coefficient of determination (R^2 LOG).

The formulas are as follows:

$$MSE = \frac{1}{N}\sum_{n=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_t^2\right)^2 \tag{7}$$

$$MAE = \frac{1}{N}\sum_{n=1}^{N}\left|\sigma_t^2 - \widehat{\sigma}_t^2\right| \tag{8}$$

$$R^2\ LOG = \frac{1}{N}\sum_{n=1}^{N}\left(ln\frac{\sigma_t^2}{\widehat{\sigma}_t^2}\right)^2 \tag{9}$$

Among them, σ_t^2 represents the squared logarithmic return of spot gold, serving as a proxy for real volatility; sigma hat sub t squared is the model's estimated forecast value.

2.2.2. Out-of-sample R^2_{oos} test

To assess the predictive performance of a model relative to a benchmark model, this study adopts the out-of-sample R^2 (R^2_{oos}) test proposed by Rapach et al. (2010). This method has been widely adopted in the literature due to its effectiveness in comparing predictive models (Liang et al., 2020; Wei et al., 2020, 2023). Since this test requires a benchmark model to be specified in advance, this study selects the classic GARCH(1,1) model as the benchmark model for the empirical evaluation, measuring the relative predictive capability of GARCH-MIDAS-type models.

The formula for out-of-sample R^2 is as follows:

$$R^2_{oos} = 1 - \frac{\sum_{t=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_{t,i}^2\right)^2}{\sum_{t=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_{t,benchmark}^2\right)^2} \tag{10}$$

Among these, σ_t^2 is a proxy for the true volatility, and this study uses the square of the logarithmic returns of gold spot prices as an approximation; $\widehat{\sigma}_t^2$ denotes the predicted volatility of prediction model i; and $\widehat{\sigma}_{t,benchmark}^2$ represents the predicted volatility of the benchmark model, GARCH(1,1).

This test method draws on the research framework proposed by Clark and West (2007), whose null hypothesis states that the mean squared error (MSE) of the benchmark model is no greater than the MSE of prediction model i. Therefore, when the R^2_{oos} value of a model is positive and statistically significant in rejecting the null hypothesis, it indicates that the prediction error of that model is significantly lower than that of the benchmark model, thereby demonstrating superior out-of-sample prediction performance.

2.2.3. Diebold and Mariano (DM) test

In addition, the predictive accuracy comparison test method proposed by Diebold and Mariano (2002) (DM test) is also widely adopted in the current literature (Mariano & Preve, 2012; Wang et al., 2020; Welch & Goyal, 2008). The null hypothesis of the DM test is that two predictive models have statistically equivalent predictive capabilities. Therefore, when the p-value obtained from the test is less than the pre-set significance level, the null hypothesis can be rejected, indicating a statistically significant difference in predictive performance between the two models. In this study, the GARCH(1,1) model remains the benchmark for comparison and analysis.

The statistical quantity of the DM test is calculated as follows:

$$DM = \frac{\overline{d}}{\sqrt{var(d_s)}} \sim N(0,1) \tag{11}$$

Among them,

$$d_s = E_{a,s}^2 - E_{b,s}^2 \tag{12}$$

$E_{a,s}$ and $E_{b,s}$ represent the prediction errors of models a and b at time s, respectively. $\overline{d}$ is the sample mean of the error difference sequence d_s, and $var(d_s)$ is its variance estimate.

Furthermore, let α_1 denote the loss function value of the extended GARCH-MIDAS model, and α_2 denote the loss function value of the traditional GARCH(1,1) model. The loss ratio between the two models can be expressed as $\frac{\alpha_1}{\alpha_2}$. If this ratio is less than 1 and the corresponding DM test statistic is significantly negative, the prediction model is statistically superior to the benchmark model. Additionally, this ratio can be used to measure the improvement in prediction loss achieved by the improved model relative to the benchmark model.

3. Data sources

The daily spot price data for gold used in this study are sourced from the Investing.com data platform.[1] The U.S.-China Tension Index (UCT) is cited from the research of Rogers et al. (2024), and the relevant index data can be obtained free of charge from the Economic Policy Uncertainty website.[2] Based on data availability and alignment with the research objectives, the sample period is set from January 1, 1993, to February 29, 2024.

To measure the volatility of gold spot returns, this study uses daily logarithmic returns as a proxy for volatility and standardizes them by multiplying by 100. Similarly, the UCT index is calculated by taking the natural logarithm, calculating its first-order difference, and multiplying by 100 to represent its monthly rate of change. Fig. 1 illustrates the time series dynamics of the daily spot gold price and the UCT index. The figure shows that during several major geopolitical or financial events, such as the 2008 global financial crisis, the peak of the U.S.-China trade war in 2018–2020, and the new round of tensions after 2022, the UCT index and the price of gold exhibited a notable correlation. During these periods, the rise in global systemic risks heightened market risk aversion, driving gold prices upward in tandem, as gold's high sensitivity to geopolitical conflicts reflects its status as a safe-haven asset.

Table 1 presents the descriptive statistical characteristics. Firstly, the average daily return of spot gold is 0.022, indicating that it exhibits neutral long-term volatility. During the sample period, the maximum and minimum daily returns were 10.437 and −8.876, respectively, suggesting that the market may experience significant volatility under specific circumstances. The yield skewness is −0.054, indicating slight left skew, and the kurtosis is 7.970, which significantly exceeds the theoretical value for a normal distribution. This indicates that the yield distribution exhibits pronounced peaks and heavy tails, consistent with the non-normal distributions commonly observed in financial assets.

Moreover, the mean of RV is 21.116, reflecting significant volatility in the gold market during the sample period. The maximum value is 195.909, while the minimum is 0.773, indicating that the market may experience severe volatility in response to major event shocks. The skewness of RV is 3.708, and the kurtosis reaches 18.513, both indicating that the RV distribution exhibits pronounced right-skewed and heavy-tailed characteristics.

Similarly, the UCT index's standard deviation is 18.811, indicating strong volatility. Its skewness is −0.012, close to symmetry but slightly left-skewed; the kurtosis is 2.695, slightly below the normal distribution level, indicating a relatively flat distribution.

After the Augmented Dickey-Fuller (ADF) test, all sequences passed the stationarity test at the 1 % significance level, meeting the requirements for subsequent time-series modeling. Additionally, the Jarque-Bera test results rejected the normality assumption for all three datasets, further validating the common skewness and heavy-tailed distribution characteristics of financial time series.

4. Empirical results

4.1. In-sample estimation results analysis

Table 2 reports the parameter estimates for each model within the sample. The results show that all parameters used to characterize short-term volatility behavior satisfy the conditions: $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta < 1$, and are significant at the 1 % significance level. This indicates that the spot gold market exhibits significant volatility clustering characteristics, i.e., volatility tends to persist and reinforce itself.

The focus of this study is to explore the impact of exogenous variables on the long-term volatility of the spot gold market, with particular attention to parameter θ. Empirical results indicate that the coefficient θ_{RV} of realized volatility is significantly positive, suggesting that monthly realized volatility has a positive impact on the long-term volatility of gold returns. Furthermore, the coefficient θ_{UCT} of UCT is also positive and significant, confirming that geopolitical risk has significant explanatory power for the long-term volatility of gold. In other words, increased tensions in U.S.-China relations will significantly increase the long-term variance of gold

[1] Investing.com Platform: https://www.investing.com/currencies/xau-usd-historical-data.
[2] Economic Policy Uncertainty Index website: https://policyuncertainty.com/US_China_Tension.html.

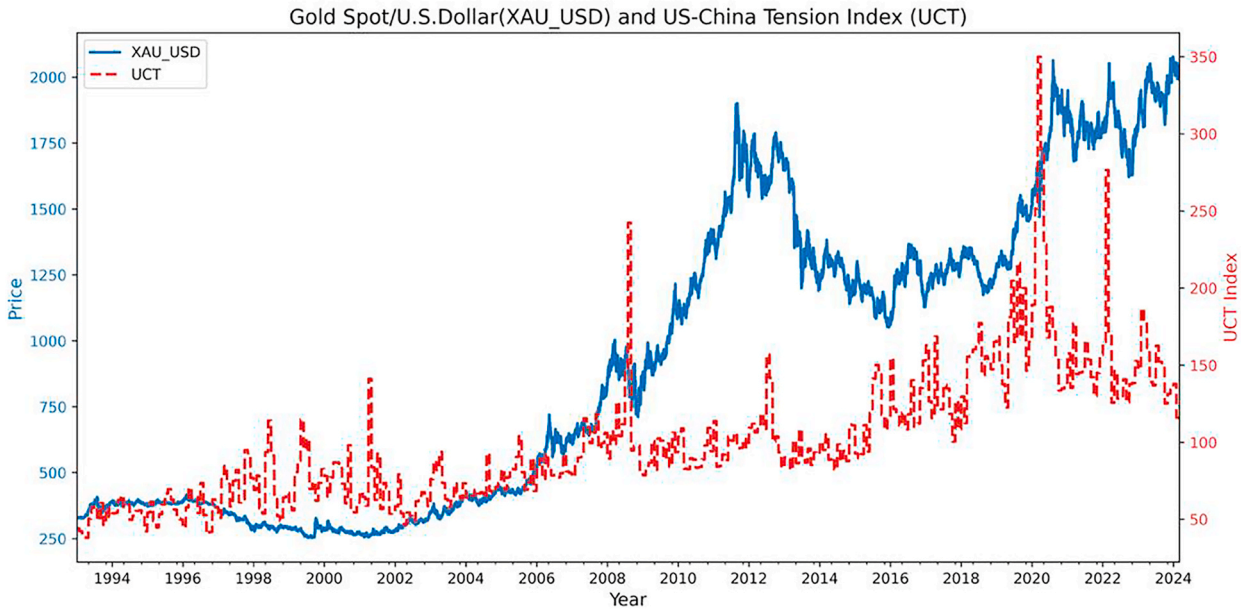

Fig. 1. Spot gold price movements and the U.S.-China tension index (UCT).

Table 1
Descriptive statistics.

	Mean	Max	Min	Std	Skew.	Kurt.	ADF	J-B
XAU_USD	0.022	10.437	−8.876	0.986	−0.054	7.970	−34.906***	21446.562***
RV	21.116	195.909	0.773	24.233	3.708	18.513	−5.348***	6040.693***
UCT	0.376	87.142	−93.971	18.811	−0.012	2.695	−9.656***	108.864***

Note: J-B stands for Jarque-Bera test for normal distribution, whose original hypothesis is that the series obeys normal distribution; ADF stands for Augmented Dickey-Fuller unit root test, whose null hypothesis is that the series contains a unit root, i.e., is non-stationary. ***,**,* denote the rejection of the original hypothesis at the 1 %, 5 % and 10 % significance levels, respectively.

returns.

This evidence demonstrates several underlying economic and behavioral mechanisms linking U.S.–China tensions to gold's long-term volatility. Firstly, intensifying U.S.–China strategic friction has accelerated the global trend of reserve diversification and partial "de-dollarization", prompting central banks to increase their gold allocations. Heightened concerns over potential financial sanctions, reserve confiscation, and geopolitical fragmentation have reinforced gold's role as an ultimate reserve asset with perceived "sanctions-immunity" characteristics (Arslanalp et al., 2022; Ngo et al., 2024). Sustained official-sector purchases, combined with the inherently low short-run supply elasticity of gold, raise the asset's long-term price sensitivity and amplify its volatility.

Secondly, the deterioration of bilateral relations is widely interpreted by market participants as signaling structural shifts in global

Table 2
In-sample estimates of model parameters.

	$GARCH-MIDAS_{RV}$	$GARCH-MIDAS_{UCT}$	$GARCH-MIDAS_{RV+UCT}$
μ	0.005 (0.010)	0.005 (0.009)	0.005 (0.010)
α	0.045*** (0.001)	0.041*** (0.001)	0.044*** (0.001)
β	0.948*** (0.001)	0.955*** (0.001)	0.948*** (0.002)
m	−0.508*** (0.129)	0.061 (0.143)	−0.647*** (0.126)
θ_{RV}	0.028*** (0.005)		0.031*** (0.004)
θ_{UCT}		0.132*** (0.019)	0.150*** (0.020)
$w_{2,\mathrm{RV}}$	1.000*** (0.252)		1.000*** (0.221)
$w_{2,\mathrm{UCT}}$		1.000*** (0.068)	1.000*** (0.062)
LLH	−9793.435	−9794.038	−9783.562
AIC	19598.870	19600.076	19583.124
BIC	19640.271	19641.477	19638.325

Note: This table reports the parameter estimates of the factor models specified with varying lag structures. LLH is the log-likelihood function, AIC denotes the Akaike information criterion, and BIC indicates the Bayesian information criterion. The numbers in parentheses are the standard errors. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

economic governance. This elevates global risk aversion, induces precautionary demand for safe-haven assets, and triggers institutional deleveraging. As a result, gold allocation shifts from short-term tactical positioning toward longer-horizon strategic holding (Baur & Smales, 2020). Such persistent demand pressure creates "stickiness" in safe-haven flows, deepening volatility clustering and contributing to UCT's significant incremental explanatory power even after controlling for realized volatility.

Lastly, U.S.–China tensions amplify policy-driven macroeconomic uncertainty through channels including trade conflicts, supply-chain restructuring, technological decoupling, and geopolitical instability. These developments heighten global inflation expectations and increase volatility in real interest rates. As a hard asset with minimal industrial uses and very low substitution elasticity, gold responds strongly to shifts in macroeconomic conditions induced by geopolitical shocks (Lucey et al., 2022). The continued significance of the UCT coefficient after controlling for RV thus reflects its ability to capture structural, policy-induced uncertainty that is not embedded in historical price movements.

In summary, θ_{UCT} is significantly positive, indicating that the higher the level of geopolitical tension between the U.S. and China, the greater the long-term variance in the gold market. This result not only reveals the spillover effects of macro-political uncertainty but also provides important insights for investors and policymakers: when assessing the risk characteristics of the gold market, the tense relationship between the U.S. and China should be regarded as a key source of exogenous shocks.

4.2. Out-of-sample forecast analysis

This section aims to evaluate the out-of-sample forecasting ability of the GARCH-MIDAS model. Compared to in-sample fitting accuracy, financial market participants are generally more concerned with a model's ability to predict future market volatility than its ability to explain historical data. To this end, this study selects the last 1000 trading days from the entire sample as a rolling prediction window. It employs a rolling prediction method to assess the model's out-of-sample performance, thereby better reflecting real-world application scenarios and evaluating its prediction stability and effectiveness in dynamic market environments.

4.2.1. Model Confidence Set (MCS) test

Table 3 presents the MCS test results based on three different loss functions. First, under the MSE loss function, all models passed the test. Among them, the $GARCH-MIDAS_{RV}$ model, which only introduces the RV term, performed the best, with a p-value of 1, significantly better than other models. Second, under the MAE and R^2 LOG loss functions, only the $GARCH-MIDAS_{RV+UCT}$ model remains within the MCS, indicating that this model has a significant advantage under these two loss criteria. Combining the results from the three loss functions, it is evident that within the specified out-of-sample prediction interval, the $GARCH-MIDAS_{RV+UCT}$ model, which incorporates both RV and UCT, demonstrates the best out-of-sample prediction capability.

4.2.2. Out-of-sample R^2_{oos} test

Furthermore, out-of-sample R^2_{oos} tests were conducted to compare the predictive performance of each forecast model with the benchmark model, as shown in Table 4. The empirical results indicate that only the $GARCH-MIDAS_{RV}$ and $GARCH-MIDAS_{RV+UCT}$ models have significant positive out-of-sample R^2_{oos} values, at 0.55 % and 0.26 %, respectively. This indicates that both the $GARCH-MIDAS_{RV}$ model, which incorporates only RV information, and the $GARCH-MIDAS_{RV+UCT}$ model, which incorporates both RV and UCT information, demonstrate significantly superior predictive performance compared to the benchmark GARCH(1,1) model in the out-of-sample data.

4.2.3. Diebold and Mariano (DM) test

Finally, this study employs the DM test proposed by Diebold and Mariano (2002) to statistically evaluate the predictive performance of each model relative to the benchmark model, with the results presented in Table 5. Under the MSE loss function, although the DM statistics of the three GARCH-MIDAS models were negative and the loss function ratios were less than 1, indicating that the prediction errors were lower than those of the benchmark model, none of them were statistically significant. Therefore, the null hypothesis that the two models have the same predictive ability cannot be rejected. In contrast, under the MAE and R^2 LOG loss functions, all GARCH-MIDAS models significantly outperformed the benchmark model, with the DM statistics being significantly negative. Further comparison of the loss function ratios reveals that the $GARCH-MIDAS_{RV+UCT}$ model, which incorporates both RV

Table 3
MCS test results.

1000 days	MSE		MAE		R^2 LOG	
	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
$GARCH(1,1)$	0.256	0.164	0.000	0.000	0.000	0.000
$GARCH-MIDAS_{RV}$	**1.000**	**1.000**	0.001	0.007	0.000	0.007
$GARCH-MIDAS_{UCT}$	0.556	0.356	0.002	0.007	0.000	0.007
$GARCH-MIDAS_{RV+UCT}$	0.556	0.449	**1.000**	**1.000**	**1.000**	**1.000**

Note: The numbers in the table are p-values. Bold and underlined values indicate the model achieving the highest predictive accuracy in each comparison. The time horizon is 1000 days.

Table 4
Out-of-sample R^2_{oos} tests.

1000 days	R^2_{oos}(%)	$MSFE-adjusted$	$p-value$
$GARCH-MIDAS_{RV}$	0.550**	0.030	0.027
$GARCH-MIDAS_{UCT}$	0.150	0.011	0.119
$GARCH-MIDAS_{RV+UCT}$	0.260**	0.018	0.047

Note: This table reports the out-of-sample forecasting performance based on the out-of-sample R^2 test proposed by Clark and West (2007). The null hypothesis states that the benchmark model has significantly smaller or equal mean squared error (MSE) compared to the evaluated model i. In this study, the benchmark model is specified as the GARCH(1,1) model. A positive value of out-of-sample R^2 indicates that the forecasting model of interest outperforms the benchmark model in terms of predictive accuracy. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

and UCT information, performs the best, reducing the prediction losses under MAE and R^2 *LOG* by 2.8 % and 2.5 %, respectively, which demonstrates stronger out-of-sample prediction capability and robustness.

In general, the results of the three out-of-sample forecasting evaluation methods consistently indicate that the $GARCH-MIDAS_{RV+UCT}$ model, which simultaneously incorporates RV and UCT, exhibits the best forecasting performance. Regardless of the loss function or test method used, this composite model significantly outperforms single-factor GARCH-MIDAS models that include only RV or UCT in capturing future volatility in the gold market. This finding highlights the complementary roles of RV and UCT in volatility forecasting, further validating the significant value of multi-source information fusion in enhancing prediction accuracy (D. Li, Zhang, & Li, 2023; Li et al., 2025). Therefore, incorporating multi-dimensional variables to construct composite forecasting models in financial market volatility modeling is not only reasonable but also practically necessary.

5. Robustness checks

5.1. Tests for different lags

To ensure that the benchmark results are independent of specific MIDAS lag structures, this study further conducted sensitivity tests on the model using different lag lengths. Beyond the benchmark settings, this study additionally selected lag lengths of 6 months before and after the benchmark value, namely, $K = 30$ and $K = 42$, for robustness testing.

The MCS test results in Table 6 indicate that when $K = 30$ months, the model maintains optimal predictive performance across all three loss functions (*MSE*, *MAE*, and R^2 *LOG*). When $K = 42$ months, the model still holds the optimal position under these three loss functions. This result demonstrates that the model's predictive performance across different lag period selections aligns with the primary out-of-sample results of this study (Table 3).

In addition, the out-of-sample R^2_{oos} test in Table 7 shows that when $K = 30$ months, the R^2_{oos} of the $GARCH-MIDAS_{RV+UCT}$ model is significantly positive (0.37 %), indicating that this model has the best out-of-sample predictive performance compared with the benchmark model. At the same time, when $K = 42$ months, the model still maintains a significantly positive R^2_{oos} value (0.58 %). This result is also consistent with the main out-of-sample test results (Table 4).

Finally, the DM test results are shown in Table 8. At $K = 30$ months, the $GARCH-MIDAS_{RV+UCT}$ model again maintains the optimal solution. Under the three loss functions, the prediction loss decreased by 0.4 %, 1.4 %, and 1.4 %, respectively, compared to the baseline model. Simultaneously, at $K = 42$ months, the $GARCH-MIDAS_{RV+UCT}$ model maintains the best predictive capability under both *MAE* and R^2 *LOG* loss functions. Prediction loss decreased by 1.9 % and 1.8 %, respectively. This result further demonstrates that out-of-sample validation across different lag periods aligns with the main conclusions of this study (Table 5).

5.2. Tests for different forecast periods

Given that out-of-sample prediction performance may be sensitive to the length of the prediction interval, it is necessary to redefine the out-of-sample prediction interval and re-evaluate the results to ensure the robustness of the assessment. Based on this

Table 5
Results of the DM Test between GARCH(1,1) and each Model.

1000 days	*MSE*	*MAE*	R^2 *LOG*
$GARCH-MIDAS_{RV}$	0.994 (−1.256)	0990*** (−3.071)	0.991*** (−4.517)
$GARCH-MIDAS_{UCT}$	0.999 (−0.590)	0.981*** (−8.379)	0.984*** (−9.576)
$GARCH-MIDAS_{RV+UCT}$	0.997 (−0.838)	0.972*** (−9.349)	0.975*** (−11.003)

Note: This table displays the results of Diebold-Mariano forecast evaluation for various models relative to the GARCH(1,1) model as the benchmark. Ratios less than 1 indicate that the model's forecast for volatility is superior to the benchmark model. DM test statistics are shown in parentheses. The time horizon is 1000 days. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

Table 6
MCS test results.

1000 days	*MSE*		*MAE*		R^2 *LOG*	
	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
K = 30						
$GARCH(1,1)$	0.459	0.291	0.037	0.068	0.008	0.017
$GARCH-MIDAS_{RV}$	0.853	0.853	0.018	0.214	0.004	0.058
$GARCH-MIDAS_{UCT}$	0.459	0.291	0.037	0.214	0.008	0.058
$GARCH-MIDAS_{RV+UCT}$	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
K = 42						
$GARCH(1,1)$	0.335	0.211	0.014	0.021	0.000	0.007
$GARCH-MIDAS_{RV}$	**1.000**	**1.000**	0.464	0.464	0.200	0.200
$GARCH-MIDAS_{UCT}$	0.152	0.089	0.012	0.021	0.001	0.007
$GARCH-MIDAS_{RV+UCT}$	0.977	0.977	**1.000**	**1.000**	**1.000**	**1.000**

Note: The numbers in the table are p-values. Bold and underlined values indicate the model achieving the highest predictive accuracy in each comparison. The time horizon is 1000 days.

Table 7
Out-of-sample R^2_{oos} test.

1000 days	$R^2_{oos}(\%)$	$MSFE-adjusted$	$p-value$
K = 30			
$GARCH-MIDAS_{RV}$	0.350*	0.020	0.079
$GARCH-MIDAS_{UCT}$	0.010	0.001	0.332
$GARCH-MIDAS_{RV+UCT}$	0.370**	0.023	0.030
K = 42			
$GARCH-MIDAS_{RV}$	0.580**	0.041	0.047
$GARCH-MIDAS_{UCT}$	−0.410	−0.004	0.667
$GARCH-MIDAS_{RV+UCT}$	0.580**	0.039	0.029

Note: This table reports the out-of-sample forecasting performance based on the out-of-sample R^2 test proposed by Clark and West (2007). The null hypothesis states that the benchmark model has significantly smaller or equal mean squared error (MSE) compared to the evaluated model i. In this study, the benchmark model is specified as the GJR-GARCH(1,1) model. A positive value of out-of-sample R^2 indicates that the forecasting model of interest outperforms the benchmark model in terms of predictive accuracy. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

consideration, this study adjusts the out-of-sample prediction interval to 500 trading days, in line with the original setting, constructs a new rolling prediction window, and re-examines the out-of-sample prediction capabilities of each model.

As shown in Table 9, the MCS test results based on the 500-day window indicate that the relative rankings of the models under the three loss functions (MAE, MSE, and R^2 *LOG*) are highly consistent with the results obtained previously using a 1000-day out-of-sample window (Table 3). This result suggests that the models' relative predictive performance is robust and consistent, further enhancing the reliability of the empirical conclusions in this study.

Additionally, as shown in Table 10, although the R^2_{oos} values of all models are positive, indicating that their predictive performance is superior to the benchmark model, only the $GARCH-MIDAS_{RV+UCT}$ model, which simultaneously incorporates RV and UCT, is

Table 8
Results of the Diebold–Mariano test between GARCH(1,1) and each Models.

1000 days	*MSE*	*MAE*	R^2 *LOG*
K = 30			
$GARCH-MIDAS_{RV}$	0.997 (−0.863)	0.994** (−2.094)	0.993*** (−4.240)
$GARCH-MIDAS_{UCT}$	0.999 (−0.153)	0.996*** (−4.264)	0.997*** (−4.544)
$GARCH-MIDAS_{RV+UCT}$	0.996 (−1.129)	0.986*** (−4.544)	0.986*** (−7.178)
K = 42			
$GARCH-MIDAS_{RV}$	0.994 (−0.802)	0.985*** (−3.390)	0.986*** (−6.382)
$GARCH-MIDAS_{UCT}$	1.004 (1.418)	1.010 (3.185)	1.007 (3.025)
$GARCH-MIDAS_{RV+UCT}$	0.994 (−0.933)	0.981*** (−4.362)	0.982*** (−6.939)

Note: This table displays the results of Diebold-Mariano forecast evaluation for various models relative to the GARCH(1,1) model as the benchmark. Ratios less than 1 indicate that the model's forecast for volatility is superior to the benchmark model. DM test statistics are shown in parentheses. The time horizon is 1000 days. ***, **, * represent rejecting the null hypothesis at the 1 %, 5 %, and 10 % significance levels, respectively.

statistically significant, with an R^2_{oos} value of 0.78 %. This result is consistent with the test results under the 1000-day out-of-sample interval setting (Table 4), further indicating that the model possesses strong stability and predictive advantages in integrating short-term (RV) and long-term (UCT) information.

Furthermore, the results of the DM test, as shown in Table 11, also support the above conclusion. Under the MAE and R^2 *LOG* loss functions, the $GARCH-MIDAS_{RV+UCT}$ model demonstrates significantly better prediction performance than other comparison models, with prediction losses reduced by 3.2 % and 2.9 %, respectively. This result once again confirms that incorporating both high-frequency volatility information (RV) and geopolitical risk factors (UCT) into the model helps to more comprehensively capture the structural characteristics of market volatility, thereby significantly improving prediction accuracy. This further highlights the theoretical significance and practical value of integrating multi-dimensional information variables for financial market volatility modeling and prediction.

Taken together, across the two aforementioned lag structures and different forecasting periods, models incorporating both realized volatility (RV) and US-China tensions (UCT) consistently demonstrate superior predictive performance. This finding indicates that the model's incremental forecasting ability is independent of specific lag periods or sample divisions, further validating the robustness and universality of this paper's benchmark conclusions.

6. Conclusions and implications

6.1. Conclusions

This study extends the GARCH-MIDAS model proposed by Engle et al. (2013) and introduces the U.S.–China Tension Index (UCT) as an exogenous predictive variable for the first time, systematically examining its impact on spot gold price volatility. The empirical results show that: (1) the spot gold market exhibits significant volatility clustering characteristics, confirming the high sensitivity of gold assets in a highly uncertain environment; (2) the UCT has a significant amplifying effect on the long-term volatility of gold prices, i.e., rising tensions in U.S.-China relations significantly exacerbate long-term volatility in the gold market; (3) After introducing realized volatility (RV) as a control variable, the UCT remains statistically significant, indicating that it successfully captures new information beyond traditional volatility indicators; (4) Out-of-sample forecasting tests further validate the superiority and robustness of the $GARCH-MIDAS_{RV+UCT}$ model in predicting long-term gold price volatility.

6.2. Discussions

Based on the GARCH-MIDAS model, this study presents a new framework that integrates geopolitical risk indicators and highlights the independent role of structural non-economic shocks in shaping long-term gold volatility. Unlike previous studies that mainly focused on macroeconomic variables or internal market volatility characteristics, this study introduces the UCT into the model as an exogenous variable, providing a new path for describing how geopolitical factors permeate the volatility structure of financial markets.

Empirical results show that UCT's amplification effect on long-term gold volatility is statistically significant and economically intuitive. When U.S.–China relations deteriorate, investors anticipate greater uncertainty in global trade, capital flows, and policy coordination, which increases required risk premiums and drives risk-averse capital toward safe-haven assets such as gold. In contrast to short-term volatility driven by liquidity or technical factors, UCT reflects persistent systemic tensions. The resulting asset repricing process exhibits slow-moving characteristics, giving UCT stronger explanatory and predictive power relative to short-term market noise.

Furthermore, UCT remains significant even after controlling for RV, further confirming that geopolitical risk variables do not merely capture existing market information but also provide important exogenous signals from outside the market (Li & Liu, 2025). The existence of this information gain challenges the "efficient market hypothesis" in financial economics to some extent, revealing that the market's response to sudden or structural risks may exhibit lag or incomplete absorption (Brooks et al., 2003; Dessaint & Matray, 2017; Karavias et al., 2023).

From a methodological perspective, the GARCH-MIDAS framework facilitates the effective integration of variables with varying frequencies and properties, enabling researchers to simultaneously identify both short-term drivers and long-term structural causes of market volatility. The modeling approach of this study demonstrates that incorporating geopolitical tension indices into traditional

Table 9
MCS test results.

500 days	*MSE*		*MAE*		R^2 *LOG*	
	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
$GARCH(1,1)$	0.490	0.527	0.008	0.012	0.000	0.000
$GARCH-MIDAS_{RV}$	**1.000**	**1.000**	0.026	0.038	0.001	0.005
$GARCH-MIDAS_{UCT}$	0.798	0.556	0.027	0.038	0.002	0.005
$GARCH-MIDAS_{RV+UCT}$	0.798	0.624	**1.000**	**1.000**	**1.000**	**1.000**

Note: The numbers in the table are p-values. Bold and underlined values indicate the model achieving the highest predictive accuracy in each comparison. The time horizon is 500 days.

Table 10
Out-of-sample R^2_{oos} tests.

500 days	$R^2_{oos}(\%)$	$MSFE-adjusted$	$p-value$
$GARCH-MIDAS_{RV}$	0.100	0.011	0.234
$GARCH-MIDAS_{UCT}$	0.050	0.007	0.282
$GARCH-MIDAS_{RV+UCT}$	0.780*	0.021	0.054

Note: This table reports the out-of-sample forecasting performance based on the out-of-sample R^2 test proposed by Clark and West (2007). The null hypothesis states that the benchmark model has significantly smaller or equal mean squared error (MSE) compared to the evaluated model i. In this study, the benchmark model is specified as the GARCH(1,1) model. A positive value of out-of-sample R^2 indicates that the forecasting model of interest outperforms the benchmark model in terms of predictive accuracy. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

Table 11
Results of the DM Test between GARCH(1,1) and each Model.

500 days	MSE	MAE	R^2 LOG
$GARCH-MIDAS_{RV}$	0.992 (−1.078)	0997 (−0.602)	0.999 (−0.399)
$GARCH-MIDAS_{UCT}$	1.000 (−0.072)	0.976*** (−6.012)	0.980*** (−7.502)
$GARCH-MIDAS_{RV+UCT}$	0.999 (−0.123)	0.968*** (−6.566)	0.971*** (−8.979)

Note: This table displays the results of Diebold-Mariano forecast evaluation for various models relative to the GARCH(1,1) model as the benchmark. Ratios less than 1 indicate that the model's forecast for volatility is superior to the benchmark model. DM test statistics are shown in parentheses. The time horizon is 500 days. ***, **, and * denote respectively the significance levels at 1 %, 5 %, and 10 %.

volatility modeling frameworks can effectively broaden the perspective and depth of asset volatility forecasting, thereby enhancing model stability in extreme risk environments. This finding is not only statistically significant but also provides new insights for risk management, policy responses, and asset allocation.

This result also responds to recent research discussions on the trend of "financialization of non-economic variables", which suggests that as the global risk landscape expands, factors such as diplomatic friction, policy uncertainty, and institutional conflicts are increasingly becoming important determinants of financial asset price behavior (Bansal et al., 2009; Battiston et al., 2016; Chen et al., 2023). Therefore, future volatility modeling and financial forecasting frameworks should place greater emphasis on incorporating structural risk variables to adapt to the increasingly complex global financial risk structure.

6.3. Implications

The findings of this study demonstrate that bilateral geopolitical tensions, captured by the U.S.–China Tension Index (UCT), have a significant and independent impact on the long-term volatility of the gold market. This provides important practical implications for macroprudential supervision, systemic risk monitoring, and cross-market policy responses.

Firstly, the results suggest that regulators should incorporate gold market indicators into their systemic risk monitoring dashboards. Given that gold volatility reacts sensitively to geopolitical shocks, long-term MIDAS volatility, extreme-percentile Value-at-Risk (VaR), and volatility jump frequency can serve as high-frequency monitoring components. Integrating these indicators into existing financial stability dashboards enables authorities to detect early signals of geopolitical risk transmission, supplementing traditional macro-financial indicators that may respond more slowly to structural political shocks.

Secondly, gold-based indicators can be operationalized within macroprudential stress-testing frameworks. UCT-driven long-term gold volatility reflects persistent geopolitical uncertainty and thus provides a valuable basis for constructing stress scenarios that account for cross-market contagion. Incorporating gold-linked shocks into stress-testing designs enables regulators to assess system-wide vulnerabilities under adverse geopolitical events better, thereby improving the forward-looking and counterfactual effectiveness of macroprudential stress tests.

Thirdly, gold market signals can support cross-market policy coordination and countercyclical regulation. When UCT intensity or long-term gold volatility exceeds threshold levels, these indicators may serve as supplementary triggers for adjusting countercyclical capital buffers, foreign exchange risk reserve ratios, or liquidity management tools. Because geopolitical tensions often lead to abrupt shifts in global safe-haven demand and cross-border capital flows, regulators can use gold-based signals to pre-emptively calibrate macroprudential measures, enhancing the resilience of the financial system.

Lastly, from a market microstructure perspective, the high sensitivity of gold prices to non-economic shocks suggests the need for enhanced real-time surveillance of trading activity. By monitoring indicators such as order-book depth anomalies, deviations in cross-market spreads, and leverage concentration in gold-related instruments, regulators can better prevent liquidity dry-ups or price dislocations during episodes of geopolitical turmoil. Temporary stabilization measures, such as strengthening margin requirements or limiting excessively leveraged speculative positions, may be considered during extreme events to prevent risk spillovers to broader financial markets.

In summary, integrating gold-based indicators into macroprudential monitoring not only enriches existing systemic risk assessment frameworks but also provides regulators with timely and geopolitically sensitive signals. Given the growing role of political risk in shaping global market dynamics, these policy pathways offer practical, effective mechanisms to strengthen early warning capabilities and enhance the robustness of financial regulation.

6.4. Shortcomings and perspectives

This study validates the UCT indicator's significant role in explaining long-term fluctuations in the gold market, though its application still has room for moderate refinement. Firstly, while UCT effectively reflects directional shifts in U.S.–China strategic friction, the risk information it captures is primarily concentrated on the bilateral dimension. This differs from traditional GPR indicators, which emphasize global political uncertainty, leading to incomplete alignment in risk coverage. Currently, there is a lack of bilateral GPR or EPU data constructed with the same logic as UCT, which limits the comparability of geopolitical risk effects from different sources. Obtaining structurally consistent bilateral political uncertainty indicators in the future would help more clearly identify the marginal impact of U.S.–China relationship shocks within a multi-layered geopolitical risk context, thereby enhancing the robustness of related conclusions.

Additionally, as UCT is extracted from public news texts, its algorithmic mechanism may still have a limited capacity to capture implicit policy signals, informal communications, or strategic interactions with lower transparency. Future research could integrate high-frequency texts, market expectation indicators, or machine learning techniques to develop more sensitive, forward-looking bilateral risk measurement tools, thereby enhancing the ability to characterize dynamic risk shifts.

Lastly, long-term gold volatility exhibits characteristics driven by multiple interrelated factors. Macro-financial variables such as US dollar liquidity, global interest rate policies, and shifts in commodity supply and demand may interact with geopolitical risks. Subsequent research could incorporate additional categories of exogenous variables within the MIDAS framework to deepen understanding of the formation mechanisms underlying long-term gold volatility.

Author contributions

S. Li: Data curation, Methodology, Writing – original draft; Y. Liu: Data curation, Validation, Writing – review & editing. All authors have made significant contributions to the research and agree to be fully accountable for the work presented in this manuscript.

AI software usage

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Data availability

The data presented in this study are available on request from the corresponding author.

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